CIRRHOSIS

Normal liver functions:
- Makes bile → help digest fat + ADEK
- Remove + ~~re~~conjugate bilirubin
- Make clotting factors
- protein synthesis
- Glycogen storage
- Fe + B12 storage
- Estrogen + aldosterone inactivation
- Detox/remove harmful substances
- Lots of blood flow!

① something causes damage to liver (ETOH, biliary problems, hepatotoxins, hepatitis B/C)
↓
② inflammation progresses and causes necrosis (liver cells unable to regenera
↓
③ permanent liver damage
↓
④ Fibrosis/scar tissue → ⑤ liver can't function normally ↓

Bile → cant make bile → cant digest fat → steatorrhea
cant digest/absorb ADEK → ↓ADEK

Clotting Factors → cant make clotting factors (↑PT/INR, ↑PTT, ↓PLT) → bleeding → petechiae, ecchymosis

Bilirubin → cant remove or conjugate (put in bile) → goes into blood (↑bilirubin) → jaundice, pruritus
→ excreted by kidneys (not stool) → clay colored stools, dark urine

Proteins → ↓protein synthesis (↓albumin)
↓
↓oncotic pressure → ascites → ↑resp infection risk
↓
peripheral edema
↓
↑skin infection risk

Estrogen → ↑estrogen → spider nevi, gynecomastia, irregular menses

Aldosterone → ↑aldosterone
↓
↑Na+H2O reabsorption in kidneys → ↑blood volume (hypervolemia) → ↑hydrostatic pressure → peripheral edem

Glycogen → cant store glycogen or do glycogenolysis → ↓Blood sugar (glucose)

Blood flow
nutrients taken to liver by portal veins and scar tissue impairs good blood flow thru liver → backup of blood in portal system → portal HTN
↓ Ascites
↓ esophageal varices → GIB, hemorrhage
→ sple meg ane

Fe/B12 → cant store → anemia

Detox → ↑toxins build up (ammonia)
↓
hepatic encephalopathy → confusion, asterixis, seizures, coma

te)

a

no-
aly
↓
mia